Importance of Custom SQL in Tableau

Scenario 1. Union of data

Tableau support four types of joins: **inner, left, right, full outer**. But this is joining two or more datasets horizontally. Tableau natively doesn't support UNION statement. UNION means vertically joining two or more datasets. Example: we have Excel file with multiple sheets, where each sheet contains same data for different years. Using UNION we can combine those sheets into one dataset. Number of columns and data type in each sheet must be same for applying UNION. SQL syntax for UNION is not complex and it looks something like this:

*SELECT * FROM Sheet1 UNION (ALL) SELECT * FROM Sheet2*

We use UNION if we want to discard duplicates, and UNION ALL for keeping all records.

Sheet1			Sheet2			UNION	
ID	Value		ID	Value		ID	Value
ID1	4		ID1	4		ID1	4
ID2	5		ID2	5		ID2	5
ID3	6		ID3	6		ID3	6
ID4	7		ID4	7		ID4	7
						ID1	4
						ID2	5
						ID3	6
						ID4	7

Scenario 2.) Tableau + Stored Procedures

Tableau has nice calculation syntax which work very nice on large dataset, but sometimes we have to apply complex logic using Table Calculations, which are prone to give bad performance on large number of records, because they are calculating in Tableau in-memory engine and not in database. So, sometimes it is better to write optimized stored procedure in database and call it inside Tableau, than writing complex calculations and making complex join logic inside Tableau.

Scenario 3.) Proper format for analysis

Sometimes data is not in proper format for data analysis. For example in Superstore Sales dataset we have Order Date and Ship Date in one row, but this is not common scenario in reality. In reality we have one one row for start date and one row for end date for same Order. It's not easy to solve this problem in Tableau, so we have to write Custom SQL. Yesterday I read an interesting article about applying Custom SQL in Tableau, so I recommend it to you.

Scenario 4.) Knowledge SQL will help you understand Tableau calculations

If you are familiar with SQL in any database, then logic behind it is something that is natural to you. If you are aggregating data then you are using GROUP BY, if you want derive some column from another table then you use Subqueries and if you want to make some analytics in database then you use WINDOW functions (OVER (PARTITION). SQL users also understand row calculations, aggregate calculations and table calculations. So it's nice to know SQL, because logic behind Tableau calcs will be easier to understand.

Like it or not, SQL will remain the best way to manipulate data. As I mentioned before Tableau is best for data visualization and advanced analytics, so we can consider SQL as complement.

What is your experience with Tableau + CustomSQL?